

1. Analysis of Audio Signal Representations for Liquid Fill-Level Estimation from Pouring Sounds

ELEC5305 Acoustics, Speech and Signal Processing (Lab Report 1)

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2. Objective

This report develops a reproducible classical signal-processing baseline for analysing water-pouring audio, supporting the proposed liquid fill-level project. Two public recordings are examined using the fast Fourier transform (FFT), short-time Fourier transform (STFT), and five frame features. Controlled experiments test window length, overlap, zero-padding, anti-alias filtering, added-noise robustness and reconstruction. The aim is to determine what these representations reveal, and which implementation choices preserve interpretable information. Absolute liquid-level prediction remains outside this report's scope because measured level labels are not used.

3. Background

Pouring audio contains cues related to container geometry and liquid level (Bagad et al., 2025), but splashes, harmonics and competing resonances can confound a spectral maximum. The FFT computes the discrete Fourier transform (DFT); a whole-record magnitude spectrum summarizes frequency content without direct temporal localization. The STFT transforms overlapping short frames; its scaled squared magnitude gives power spectral density (PSD) (MathWorks, n.d.-a, n.d.-c, n.d.-d).

Hann windowing trades reduced sidelobe leakage for a wider main lobe. Longer windows improve stationary-tone separation while averaging over longer time intervals. Zero-padding refines the FFT grid without narrowing that main lobe (Harris, 1978). Overlap changes frame spacing, not window duration. Reconstruction requires normalization by the actual accumulated analysis/synthesis weights.

Sampling below twice a component's frequency causes aliasing. A finite impulse response (FIR) low-pass filter attenuates unwanted high-frequency content before decimation (MathWorks, n.d.-b). Convolution also limits added broadband noise, although it can distort useful signal content. Signal-to-noise ratio (SNR) quantifies error against a specified reference. Root-mean-square (RMS) amplitude, spectral maximum, power centroid, spread and energy roll-off provide complementary descriptors, rather than interchangeable liquid-level measurements.

4. Methodology and Implementation

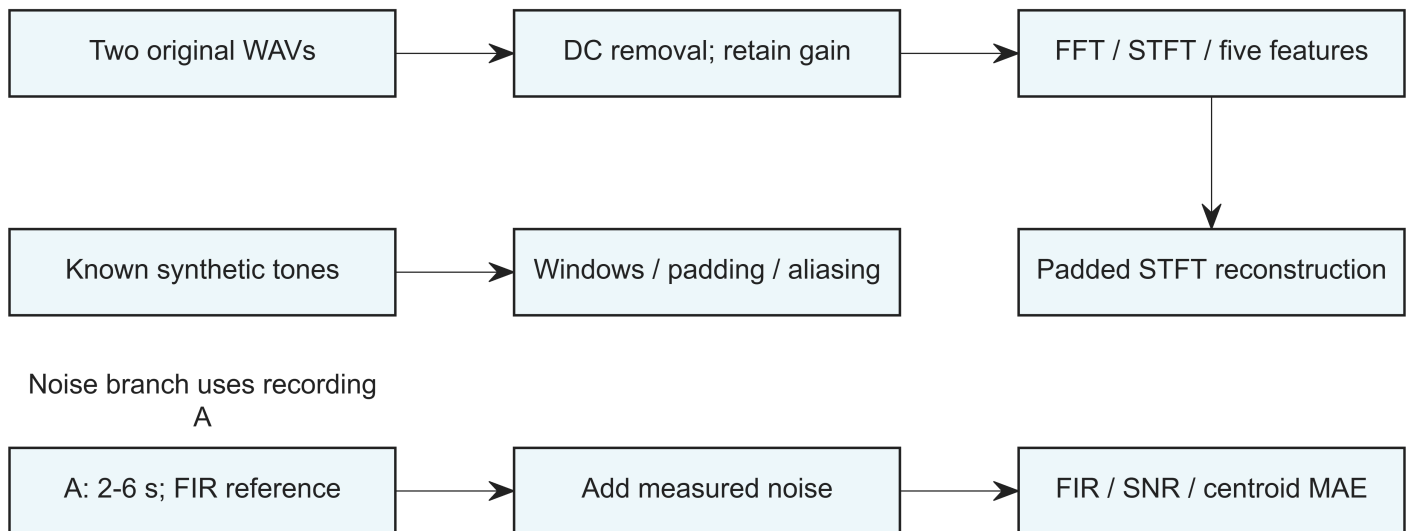
Two WAVs from the Sound of Water dataset share container_1, water, approximately constant flow and no labelled background noise (Bagad et al., n.d.). Local files pour_A.wav and pour_B.wav retain the original samples from VID_20240116_230040_2.1_16.7 and VID_20240116_230132_3.3_18.8. They contain 703488 and 745472 mono 16-bit pulse-code-modulated (PCM) samples at 48 kHz. Their measured durations are

14.656 and 15.531 s. They are already segmented; source-video timestamps are not reapplied. Each complete recording is DC-removed without amplitude normalization or silence rejection.

The pipeline branches into a whole-record Hann FFT and complete-frame STFTs. Windows of 10, 40 and 100 ms share 50% overlap, an 8192-point transform and common PSD colour limits. A separate 40 ms test uses 0%, 50% and 75% overlap. Baseline features use 40 ms and 50% overlap: spectral maxima within 300-5000 Hz; power centroid, spread and 90% roll-off within 0-8000 Hz; and unwindowed RMS. Equal-time thirds provide descriptive medians, not measured level classes.

Known-tone synthesis isolates zero-padding and aliasing. A 513-tap, 6 kHz Hamming-windowed sinc FIR filters before 48-to-16 kHz decimation. Centered convolution compensates its 256-sample delay offline. A separate 2-6 s segment of A is low-pass filtered to define a band-limited reference, then corrupted at measured 0, 10 and 20 dB SNR using five stored noise realizations. Samples within 40 ms of each boundary and frames overlapping them are excluded from evaluation. Finally, a 20 ms STFT with 50% overlap is inverted using padded, normalized overlap-add. Custom helpers keep the report readable and require only base MATLAB; optional toolbox cross-checks test equivalence.

```
ap_pipeline;
```



5. Code and Results

Run all sections in order with this report folder as MATLAB's Current Folder. The analysis uses base MATLAB; an optional Signal Processing Toolbox comparison checks the PSD. Keep code and outputs visible when exporting the report. Presentation follows the supplied ELEC5305 report instructions and three sample reports.

For frame m , frequency bin k , window length M and hop H :

$$X_m[k] = \sum_{n=0}^{M-1} x[mH + n]w[n]e^{-j2\pi kn/N_{FFT}}.$$

$$w[n] = \frac{1}{2} - \frac{1}{2} \cos(2\pi n/M), \quad H = M - O, \quad f_k = kF_s/N_{FFT}.$$

The one-sided PSD uses $dk=2$ *except at DC and Nyquist, where $dk=1$* :

$$P_m[k] = \frac{d_k |X_m[k]|^2}{F_s \sum_n w[n]^2}.$$

The spectral peak is the maximum in $B_p=300\text{-}5000$ Hz; the power-weighted centroid uses $B_c=0\text{-}8000$ Hz. RMS uses the unwrapped frame:

$$f_{p,m} = \operatorname{argmax}_{f_k \in B_p} P_m[k], \quad C_m = \frac{\sum_{f_k \in B_c} f_k P_m[k]}{\sum_{f_k \in B_c} P_m[k]}.$$

$$R_m = \sqrt{\frac{1}{M} \sum_{n=0}^{M-1} x[mH + n]^2}, \quad R_{dB,m} = 20 \log_{10}(\max(R_m, 10^{-12})).$$

DC means the zero-frequency component; NFFT is the transform length, O the overlap in samples, and F_s the sampling frequency. Full scale (FS) refers to the digital amplitude reference 1, not calibrated sound pressure.

5.1. Input checks and whole-record analysis

```
clearvars;
set(groot, 'defaultAxesFontSize', 11, 'defaultTextFontSize', 11);
files={'pour_A.wav', 'pour_B.wav'};
signals=cell(1,2); durations=zeros(1,2);
for j=1:2
    [raw,fsj]=audioread(files{j});
    assert(fsj==48000 && all(isfinite(raw(:))), 'Unexpected input format. ');
    x=mean(raw,2); % Preserve mono; safely average stereo.
    signals{j}=x-mean(x); % Remove DC without changing gain.
    durations(j)=numel(x)/fsj;
    fprintf('%c: Fs=%d Hz, N=%d, T=%.6f s, peak=%.6f FS\n', ...
        'A'+j-1,fsj,numel(x),durations(j),max(abs(raw(:))));
end
```

```
A: Fs=48000 Hz, N=703488, T=14.656000 s, peak=0.300354 FS
B: Fs=48000 Hz, N=745472, T=15.530667 s, peak=0.900177 FS
```

```
fs=48000;
ap_plot('wave_fft',signals,fs);
```

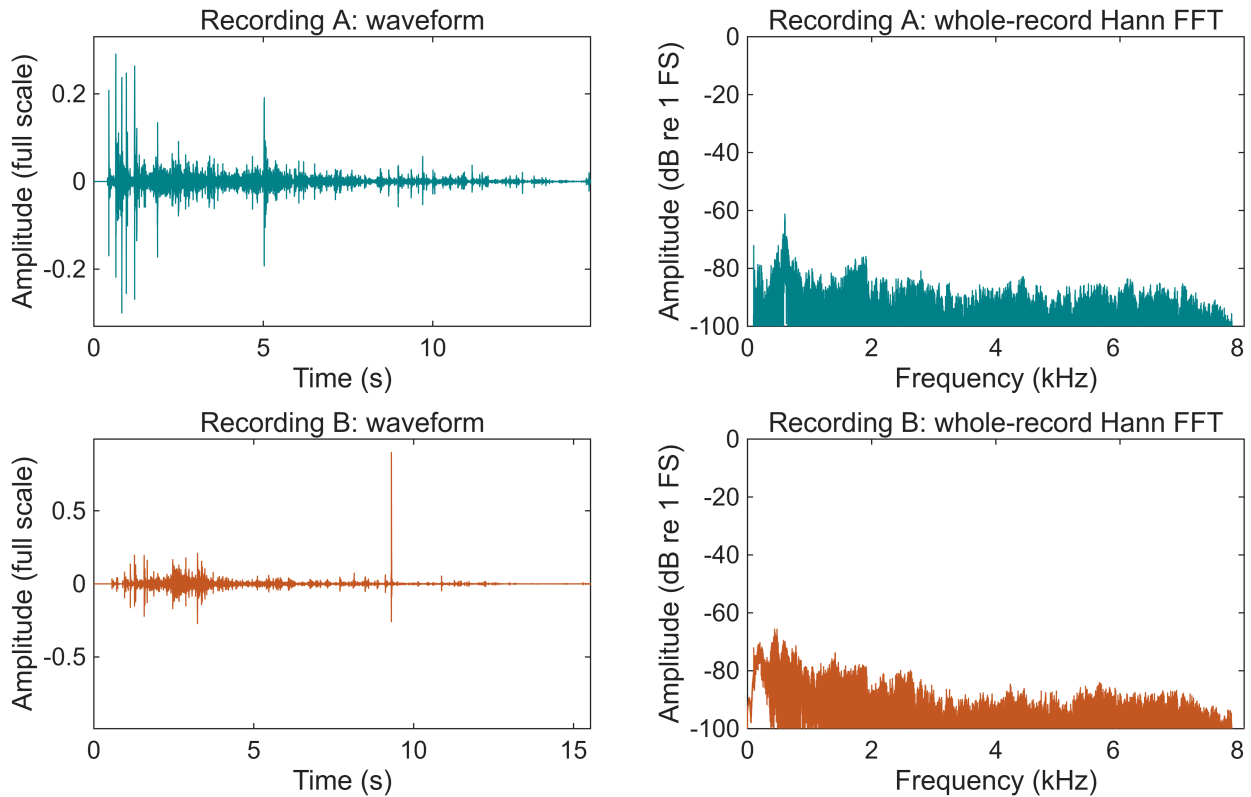


Figure 1. Complete waveforms and coherent-gain-corrected one-sided Hann FFTs for recordings A and B. Amplitude references are digital full scale. The waveform panels use different vertical limits to retain detail. The spectra contain low-frequency concentrations and broadband components but cannot localize the changes in time. B includes a strong transient near 9.3 s. Neither source file reaches a 16-bit clipping endpoint.

5.2. Window length at fixed FFT size

```

windowMs=[10 40 100]; windowRows=zeros(3,7);
for j=1:3
    b=ap_stft(signals{1},fs,windowMs(j),0.5,8192);
    windowRows(j,:)=[windowMs(j),b.M,b.H,1000*b.H/fs, ...
        b.frames,fs/8192,4*fs/b.M];
end
windowTable=array2table(windowRows,'VariableNames', ...
    {'Window_ms','M','Hop_samples','Hop_ms','Frames_A','Grid_Hz','HannWidth_Hz'});
disp(windowTable);

```

Window_ms	M	Hop_samples	Hop_ms	Frames_A	Grid_Hz	HannWidth_Hz
10	480	240	5	2930	5.8594	400
40	1920	960	20	731	5.8594	100
100	4800	2400	50	292	5.8594	40

```

ap_plot('windows',signals{1},fs);

```

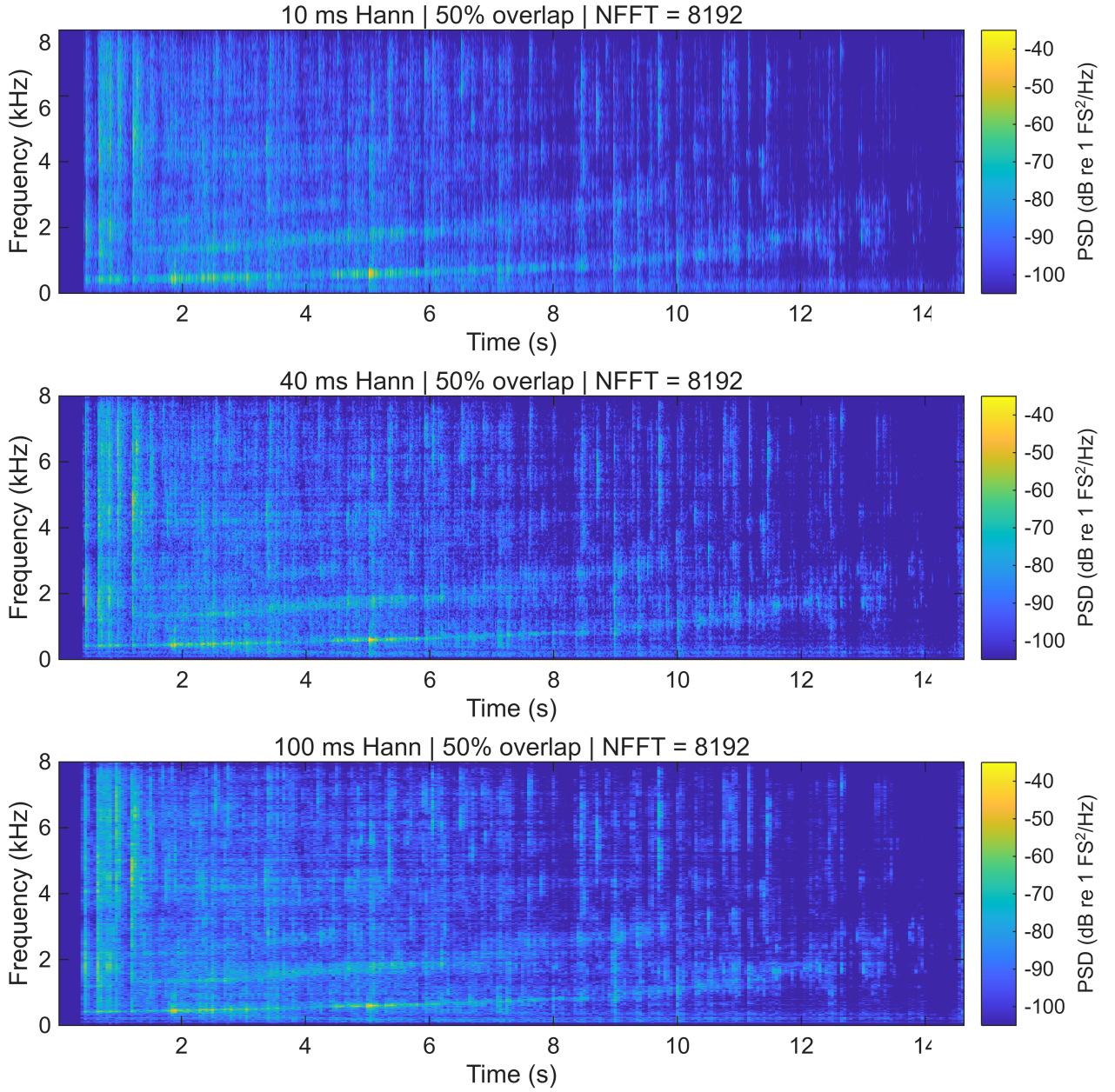


Figure 2. Recording A with 10, 40 and 100 ms periodic Hann windows. All panels use the same PSD scale (-105 to -35 dB re 1 FS²/Hz) and FFT grid. The 10 ms view emphasizes timing but broadens spectral bands. The 100 ms view sharpens sustained frequency bands while spreading short events over a longer analysis interval. The 40 ms setting is an exploratory compromise, not an accuracy-optimized choice. Table 1 reports main-lobe widths of 400, 100 and 40 Hz, although every FFT grid is 5.859375 Hz. The window/hop interpretation follows MathWorks (n.d.-d).

5.3. Frame features and repeated-recording comparison

```

base=cell(1,2); stageRows=zeros(6,5); row=0;
for j=1:2
    base{j}=ap_stft(signals{j},fs,40,0.5,8192);
    b=base{j}; tau=b.t/durations(j);
    for k=0:2
        row=row+1; mask=tau>=k/3 & tau<(k+1)/3;
        stageRows(row,:)= [j,k+1,median(b.peak(mask)), ...
            median(b.centroid(mask)),median(20*log10(max(b.rms(mask),1e-12)))];
    end
end
stageTable=array2table(stageRows,'VariableNames', ...
{'Recording_1A_2B','Third_1Early_2Middle_3Late','Peak_Hz','Centroid_Hz','RMS_dBFS'})
;
disp(stageTable);

```

Recording_1A_2B	Third_1Early_2Middle_3Late	Peak_Hz	Centroid_Hz	RMS_dBFS
1	1	503.91	1946.2	-43.008
1	2	755.86	1820	-49.399
1	3	1593.8	1832.1	-58.613
2	1	468.75	1727.4	-41.935
2	2	761.72	1656.8	-50.132
2	3	996.09	1467	-66.016

```

ap_plot('features',base,durations);

```

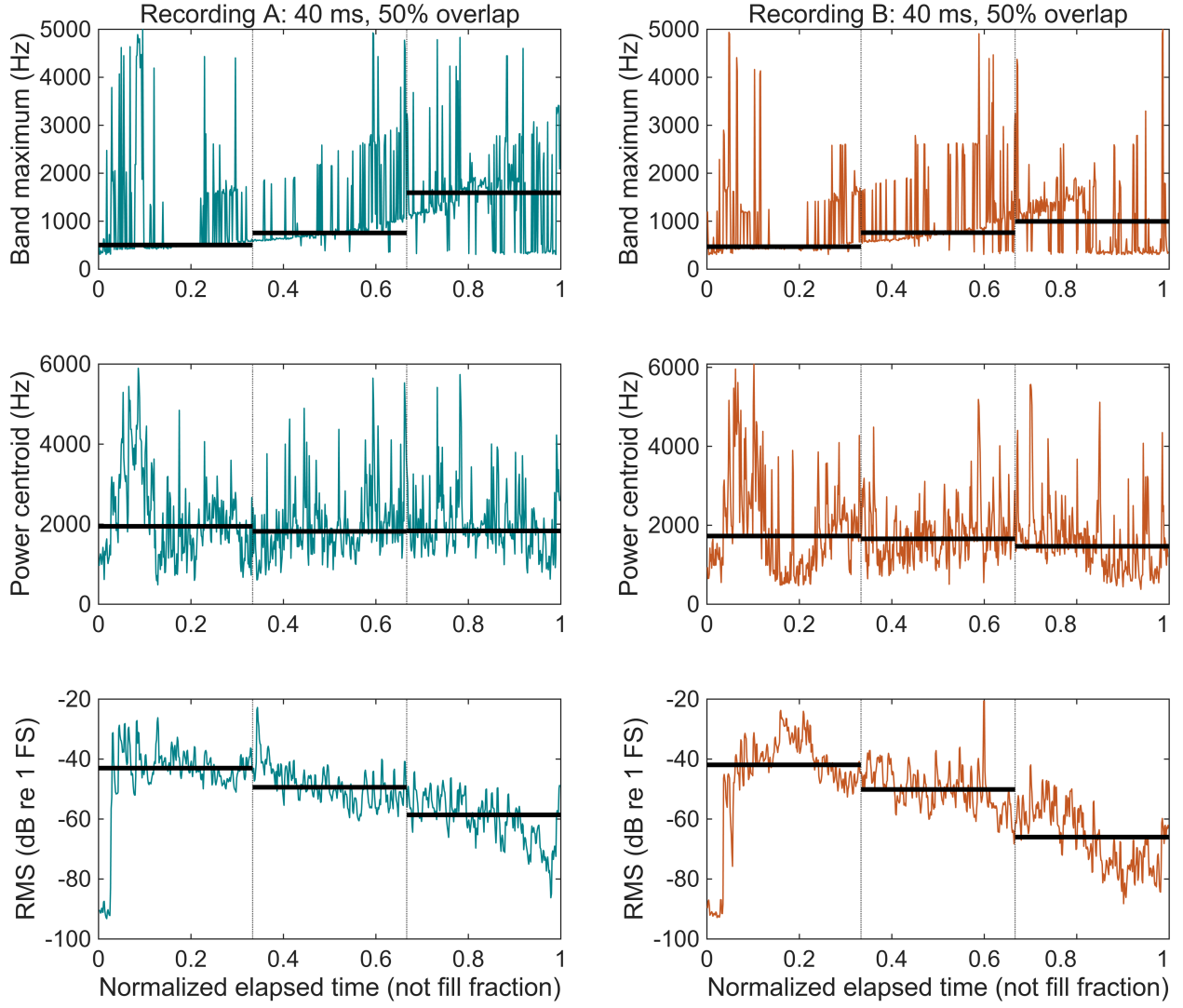


Figure 3. Frame features at 40 ms and 50% overlap. Black horizontal segments denote within-third medians; vertical dotted lines mark elapsed time thirds. The peak search is restricted to 300-5000 Hz and centroid to 0-8000 Hz. Peak medians rise from 503.9 to 1593.8 Hz for A and from 468.8 to 996.1 Hz for B. Centroids do not share this trend. RMS falls substantially toward the end. Raw trajectories retain jumps and low-energy frames so that the limitations of a simple band-maximum feature remain visible.

Additional feature definitions. With $p[k]=P[k]/\sum(P[k])$ over 0-8000 Hz, spread is the power-weighted standard deviation around centroid C :

$$B_m = \sqrt{\sum_{f_k \in B_c} (f_k - C_m)^2 p_m[k]}.$$

Roll-off is the lowest frequency below which at least 90% of band power lies:

$$R_{90,m} = \min \{f_j : \sum_{k=0}^j P_m[k] \geq 0.9 \sum_{f_k \in B_c} P_m[k]\}.$$

```
featureTable=ap_feature_summary(base,durations);
disp(featureTable);
```

Recording_1A_2B	Third	Spread_Hz	Rolloff90_Hz
1	1	1688.9	4570.3
1	2	1494.2	3861.3
1	3	1445.7	3668
2	1	1702.9	4529.3
2	2	1405.5	3023.4
2	3	1346.7	2797.9

Table 3. Additional medians in elapsed-time thirds. A's spread decreases from 1688.9 to 1445.7 Hz and 90% roll-off from 4570.3 to 3668.0 Hz. B shows the same broad direction (1702.9 to 1346.7 Hz; 4529.3 to 2797.9 Hz). These distributions narrow despite rising median band maxima; no feature is assumed to be a calibrated liquid-level measurement.

5.4. Synthetic control: window length versus zero-padding

The helper explicitly synthesizes two known cosine waves and evaluates (10 ms, NFFT=512), (10 ms, NFFT=8192) and (100 ms, NFFT=8192).

```
ap_plot('padding');
```

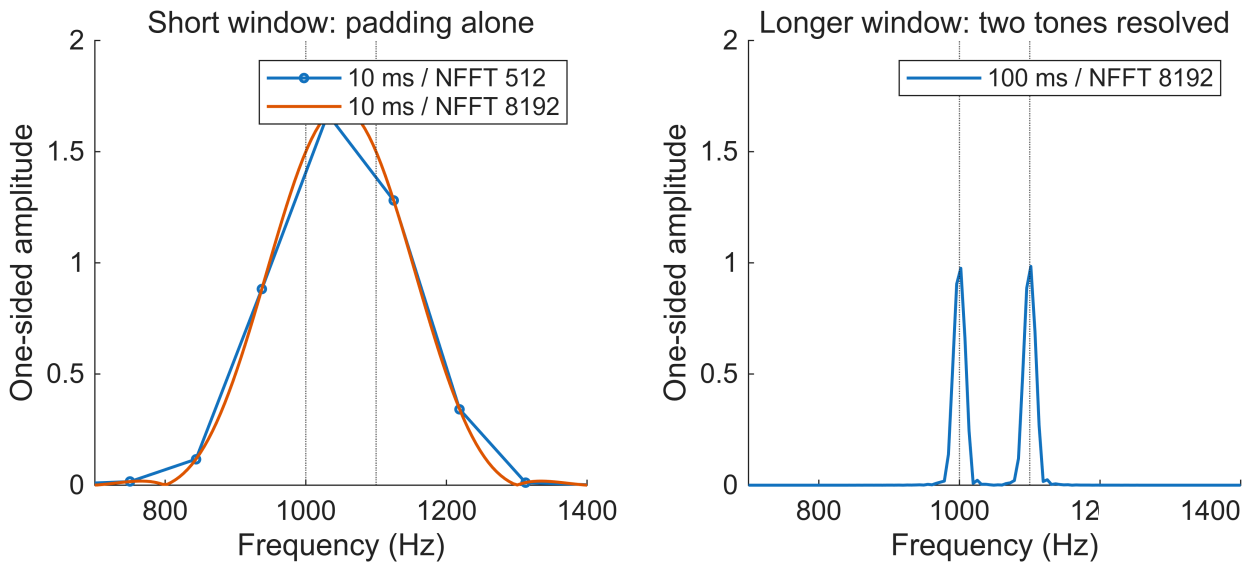


Figure 4. With 10 ms of data the two 100 Hz-separated tones form a broad merged lobe. Raising NFFT from 512 to 8192 improves sampling of that same lobe, changing grid spacing from 93.75 to 5.859375 Hz. Extending the actual window to 100 ms resolves the two known tones. This control prevents a visually smoother spectrogram being mistaken for improved resolving ability.

5.5. Overlap, scaling and reproducibility checks

```
overlap=[0 0.5 0.75]; overlapRows=zeros(3,4);
for j=1:3
    b=ap_stft(signals{1},fs,40,overlap(j),8192);
    overlapRows(j,:)= [100*overlap(j),1000*b.H/fs,b.frames, ...
        median(abs(diff(b.peak)))];
end
overlapTable=array2table(overlapRows,'VariableNames', ...
    {'Overlap_percent','Hop_ms','Frames_A','MedianAbsPeakStep_Hz'});
disp(overlapTable);
```

Overlap_percent	Hop_ms	Frames_A	MedianAbsPeakStep_Hz
0	40	366	93.75
50	20	731	52.734
75	10	1462	17.578

Table 4. At 0%, 50% and 75% overlap, A yields 366, 731 and 1462 frames. Median adjacent-frame peak changes are 93.75, 52.73 and 17.58 Hz. These changes span different time intervals and correlated frames; they are not comparable accuracy measures or evidence that 75% is intrinsically better.

Check PSD scaling, optional toolbox equivalence and reference results.

```
ap_validate(signals,base,stageRows>windowRows,overlapRows,fs);
```

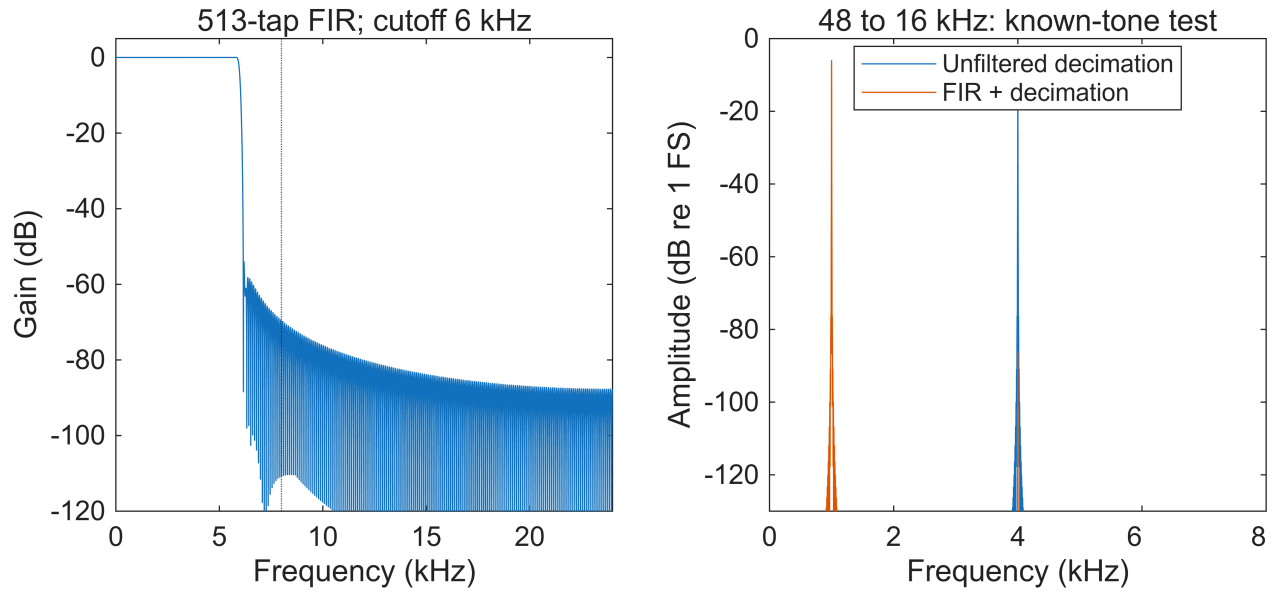
PSD scaling check passed; absolute error 1.45e-24.
spectrogram cross-check passed; relative error 6.44e-16.
All bundled-data reference checks passed.

```
if ~exist('matlab_results','dir'), mkdir('matlab_results'); end
writetable(windowTable,fullfile('matlab_results','window_parameters.csv'));
writetable(stageTable,fullfile('matlab_results','stage_results.csv'));
writetable(overlapTable,fullfile('matlab_results','overlap_results.csv'));
```

5.6. Anti-alias filtering and sampling-rate reduction

A known 1 kHz + 12 kHz cosine mixture is decimated from 48 to 16 kHz. At the output rate, 12 kHz folds to $\text{abs}(12000 - 16000) = 4000$ Hz. The FIR precedes decimation, and 50 ms boundary regions are excluded from the known-tone amplitude measurements. Its causal delay is 256/48000 s. Centered convolution compensates this delay offline; real-time use would need buffering. The same conversion also writes a correctly labelled 16 kHz version of A, without replacing the 48 kHz feature baseline.

```
rateTable=ap_ratechange(signals{1},fs);
```



```
disp(rateTable);
```

NaiveAlias_FS	FilteredAlias_FS	Retained1k_FS	AliasSuppression_dB	StopbandMaximum_dB
0.5	4.9659e-05	0.50018	80.06	-69.791

```
writetable(rateTable,fullfile('matlab_results','ratechange.csv'));
```

Figure 5 and Table 5. The FIR response and known-tone spectra show why filtering must precede decimation. Unfiltered alias amplitude is 0.5000 FS; filtered amplitude is 4.966e-5 FS, or 80.06 dB lower at this specific tone. Worst response at or above 8 kHz is -69.79 dB. The 80.06 dB number is not a universal stopband guarantee. The retained 1 kHz amplitude is 0.50018 FS.

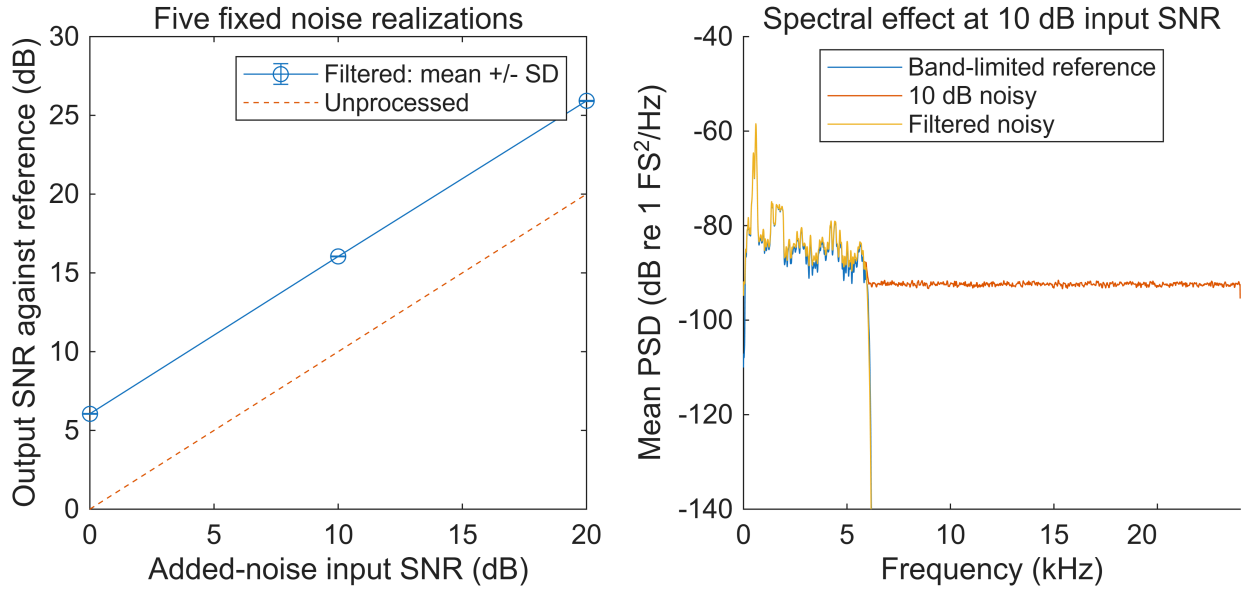
5.7. Measured additive-noise and filtering benchmark

Reference s is a low-pass-filtered segment of a real recording, not a physically clean microphone measurement. Noise is scaled using measured power on the retained interior samples; $z=s+v$ and $y=h*z$ by convolution.

$$\text{SNR}_{out} = 10 \log_{10} \frac{\sum_{n \in V} s[n]^2}{\sum_{n \in V} (y[n] - s[n])^2}.$$

Centroid MAE is the mean absolute difference from the reference's frame centroid (Hz); V is the retained sample set. Frame centres are retained only when their entire window lies inside V . Mean and sample standard deviation (SD) of output SNR use five noise realizations, not five recordings.

```
noiseTable=ap_noise(signals{1},fs);
```



```
disp(noiseTable);
```

InputSNR_dB	OutputSNR_mean_dB	OutputSNR_SD_dB	CentroidMAE_input_Hz	CentroidMAE_output_Hz
0	6.055	0.02309	1219.4	617.08
10	16.043	0.023936	307.01	136.71
20	25.921	0.026264	40.966	19.512

```
writetable(noiseTable,fullfile('matlab_results','noise_benchmark.csv'));
```

Figure 6 and Table 6. Input SNRs of 0, 10 and 20 dB yield mean output SNRs of 6.05, 16.04 and 25.92 dB. At 10 dB input, mean centroid MAE decreases from 307.0 to 136.7 Hz. A low-pass filter helps this deliberately band-limited benchmark; this does not establish general real-noise removal or improved liquid-level prediction. All operations retain the same gain.

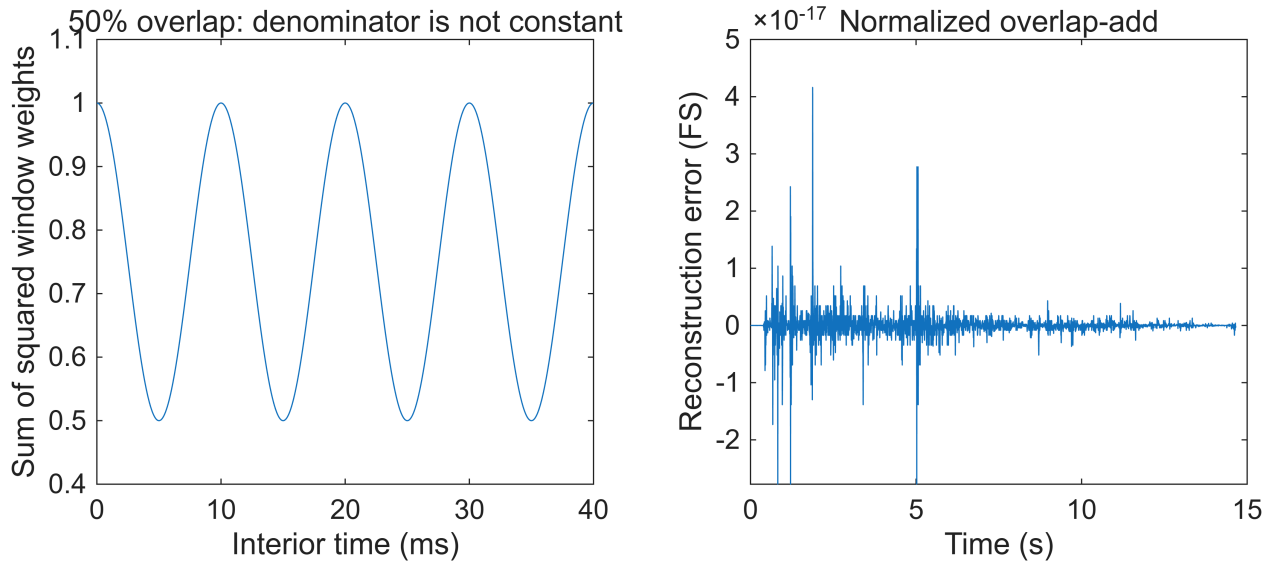
5.8. STFT reconstruction and overlap-add normalization

An inverse FFT reconstructs each frame. The synthesis window is also Hann. Normalize by the overlap of window PRODUCTS (here window squared):

$$\hat{x}[n] = \frac{\sum_m w[n - mH] u_m[n - mH]}{\sum_m w[n - mH]^2},$$

where u_m is the inverse transform of the analysis-windowed frame. Padding one window at both ends protects the original boundary samples; the final output is trimmed to the exact original length. NRMSE means normalized root-mean-square error, $\text{norm}(\hat{x}-x)/\text{norm}(x)$.

```
reconstructionTable=ap_reconstruct(signals{1},fs);
```



```
disp(reconstructionTable);
```

NRMSE	MaxAbsoluteError_FS	MinimumWeight	MaximumWeight
2.4539e-16	1.1102e-16	0.5	1

```
writetable(reconstructionTable,fullfile('matlab_results','reconstruction.csv'));
```

Figure 7 and Table 7. The interior squared-window sum varies from 0.5 to 1.0 at 50% overlap. Samplewise normalization nevertheless recovers the signal to floating-point precision (independent reference NRMSE $2.39\text{e-}16$). MATLAB need not reproduce the last numerical digits; the validation gate is $\text{NRMSE} < 1\text{e-}12$. Reconstruction verifies the transform implementation, not the correctness of acoustic peak identification.

```
writetable(featureTable,fullfile('matlab_results','additional_features.csv'));
fprintf('ALL REPORT CHECKS PASSED.\n');
```

ALL REPORT CHECKS PASSED.

6. Discussion

Band maxima increase from 503.9 to 1593.8 Hz in A and from 468.8 to 996.1 Hz in B, consistent with evolving acoustic structure (Figure 3). However, late-stage differences and framewise jumps prevent interpreting this maximum as reliable pitch. The centroid, spread and roll-off generally decrease instead (Tables 2-3): broadband energy can shift independently of a narrow resonance. Quiet frames remain visible, and robust medians limit isolated outlier influence without certifying the selected peak. The digital RMS decreases are not calibrated sound-pressure measurements.

Figures 2 and 4 distinguish observation duration from FFT grid spacing. Longer windows resolve nearby stationary tones; padding alone does not. Increasing overlap gives denser, correlated observations and greater computation, not more independent evidence. Figure 7 also distinguishes reconstruction from constant window weighting: Hann squared at 50% overlap is nonconstant, but correct samplewise normalization gives negligible reconstruction error.

The alias experiment verifies that filtering must occur before decimation (Figure 5). Suppression of the particular 12 kHz tone is 80.06 dB, whereas the worst measured stopband response is -69.79 dB. The 6 kHz cutoff deliberately sacrifices higher-frequency content and cannot preserve every possible feature.

In the constructed noise benchmark, 10 dB input becomes 16.04 dB output and centroid MAE falls from 307.0 to 136.7 Hz (Figure 6). Five realizations make comparison reproducible; their SD measures added-noise variability on one segment. Useful reference energy was deliberately restricted before noise addition, so these gains cannot be generalized to arbitrary recording noise. Filter distortion is included in output error.

The study uses two recordings from one container and no measured level labels. Future prediction requires calibrated labels, confidence-aware pitch tracking and held-out recordings/containers; random splits of overlapping frames would leak information. The results establish an analysis baseline, not classification accuracy.

7. Conclusion

This report implemented an interpretable analysis baseline for pouring audio using two real recordings, known-tone controls and a measured-noise benchmark. The FFT and STFT revealed complementary spectral information, while window and padding experiments distinguished resolving ability from display-grid density. Five features exposed both repeatable temporal changes and limitations of spectral maxima. Anti-alias filtering suppressed a specified folding tone, and controlled noise filtering improved reference-based SNR and centroid stability. Padded, normalized overlap-add recovered the original signal to floating-point precision, validating the transform mechanics. These findings meet the objective of testing representations and implementation choices for the proposed project. They do not demonstrate absolute liquid-level estimation or robustness across containers. The documented 40 ms, 50% overlap baseline is a starting point for later work with calibrated labels and reliable acoustic tracking.

8. References

- Bagad, P., Tapaswi, M., Snoek, C. G. M., & Zisserman, A. (n.d.). *Sound of Water* [Data set]. Hugging Face. Retrieved September 4, 2026, from <https://huggingface.co/datasets/bpiyush/sound-of-water>
- Bagad, P., Tapaswi, M., Snoek, C. G. M., & Zisserman, A. (2025). The sound of water: Inferring physical properties from pouring liquids. In *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 736–740). IEEE. <https://doi.org/10.1109/ICASSP49660.2025.10889950>
- Harris, F. J. (1978). On the use of windows for harmonic analysis with the discrete Fourier transform. *Proceedings of the IEEE*, 66(1), 51–83. <https://doi.org/10.1109/PROC.1978.10837>
- MathWorks. (n.d.-a). *fft*. Retrieved September 4, 2026, from <https://www.mathworks.com/help/matlab/ref/fft.html>
- MathWorks. (n.d.-b). *resample*. Retrieved September 4, 2026, from <https://www.mathworks.com/help/signal/ref/resample.html>
- MathWorks. (n.d.-c). *spectrogram*. Retrieved September 4, 2026, from <https://www.mathworks.com/help/signal/ref/spectrogram.html>
- MathWorks. (n.d.-d). *Spectrogram computation with Signal Processing Toolbox*. Retrieved September 4, 2026, from <https://www.mathworks.com/help/signal/ug/spectrogram-computation-with-signal-processing-toolbox.html>